QoS-Aware Composable Digital Twins for Resilient Coordination of Industrial Mobile Robots

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***Abstract*—Digital Twins (DTs) are key enablers for cyber-physical integration in industrial systems. However, many current implementations remain tightly coupled to specific hardware platforms and vendor ecosystems, limiting interoperability and composability. Building on a recent concept of DTs as software proxies for physical entities, we extend this concept with a QoS-aware service model and substitution mechanism that enables runtime selection among partially equivalent services, thereby supporting resilient operation through controlled degradation. The paper proposes a service-oriented architecture for proxy-based DTs tailored to autonomous industrial mobile machines, such as mining vehicles. Grounded in Eclipse Arrowhead, the architecture abstracts each machine into a secure, discoverable service encapsulating sensing and task execution capabilities. A Proof-of-Concept implementation demonstrates dynamic service registration, secure orchestration, and runtime composition of machine capabilities, including QoS-aware selection among alternative service providers under degraded conditions. The evaluation further analyses the behaviour of the QoS-aware service model under varying mission priorities and degradation conditions. The paper contributes (i) a domain-specific extension for part-autonomous industrial mobile machines, (ii) a layered composition model from machine-level iDTs to capability-level and mission-level cDTs, and (iii) a QoS-aware service model integrated into an Eclipse Arrowhead-based runtime architecture for safety-critical, connectivity-constrained environments.**

***Index Terms*—Digital Twins, Industrial IoT, Eclipse Arrowhead, Autonomous Robots, Service-Oriented Architecture**

I. Introduction

The industrial sector is undergoing rapid transformation driven by the convergence of Industrial Internet of Things (IIoT), autonomous systems, and cyber-physical production systems (CPPS) [20]. In underground mining and heavy industry, part-autonomous mobile machines such as wheel loaders, Load–Haul–Dump (LHD) vehicles, and dump trucks are increasingly equipped with advanced sensing, autonomous navigation, and tele-remote operation capabilities. These machines typically operate in safety-critical and connectivity-constrained environments, requiring robust integration between physical assets and digital infrastructures.

Digital Twins (DTs) provide virtual counterparts of physical assets, enabling monitoring, control, and optimization [6], [14]. Recent work advocates distributed and composable DT architectures, where DTs act as proxies for physical entities, enabling interoperability and modularity across systems [5]. However, current DT implementations remain tightly coupled to proprietary ecosystems [16], lack mechanisms for dynamic service discovery, orchestration [4], [19], and runtime reasoning about quality differences among functionally equivalent services. These are severe limitations in safety-critical industrial environments with intermittent connectivity [15].

Proxy-based DTs for mobile robots must overcome (i) hardware- and vendor-coupled solutions that impede interoperability and lifecycle flexibility in heterogeneous IIoT/OT environments [7], [16], (ii) the need for secure and reliable bidirectional cyber–physical interaction in safety-critical automation contexts [15], (iii) limited support for systematic composition and integration of multiple DTs into higher-level, purpose-oriented systems-of-systems at runtime [5], [17], and (iv) the lack of mechanisms to reason about quality differences among functionally equivalent services, which limits the ability to support resilience through controlled degradation. Thus, while prior work addresses DT architectures, service-oriented CPS, and QoS-aware service selection independently, existing approaches do not integrate QoS-aware service selection and failover into composable DT architectures for industrial mobile systems operating under intermittent connectivity and safety constraints.

To address this limitation, the present paper extends the proxy-based DT concept with a QoS-aware service model and substitution mechanism that enables runtime selection among partially equivalent services, thereby supporting resilient operation through controlled degradation. Compared to prior proxy-based DT work [5], this paper thus extends the approach to autonomous industrial mobile machines and integrates QoS-aware service selection into a composable architecture.

We propose a service-oriented conceptual architecture for proxy-based DTs tailored to autonomous mobile machines, following the concept of distributed DTs as proxies [5] and grounded in industrial service-oriented principles through Eclipse Arrowhead [11], [12]. In this architecture, each machine is abstracted into an individual DT (iDT) that encapsulates sensing, autonomous task execution, tele-remote control, and lifecycle management capabilities. Composite DTs (cDTs) are formed through the orchestration of iDT services within an Arrowhead local cloud, enabling layered composition from machine-level capabilities to mission-level functionality. This supports dynamic creation and recombination of services to meet varying operational needs. A Proof-of-Concept (PoC) implementation is developed to validate the architecture and its QoS-aware capabilities, i.e., proving technical viability.

The main contributions of this paper are:

- A problem analysis for part-autonomous industrial mobile machines, grounded in a representative underground mining use case, which formalizes interoperability, security, composability, and intermittent-connectivity requirements that are not addressed in the generic proxy-DT formulation of [5] (Section II).
- A QoS-aware service model for DT composition, enabling runtime selection among partially equivalent services under varying mission priorities, including a layered composition architecture (iDT–cDT) supporting dynamic integration for industrial mobile robots (Section III).
- A PoC implementation demonstrating how machines can be abstracted into Arrowhead-compliant dynamically orchestrated services, providing modularity, composability, and interoperability (Section IV).
- An evaluation, demonstrating improved resilience through controlled degradation, analysing QoS trade-offs and resilience under degradation, including failover behaviour (Section V).

In addition, the paper analyses how the Eclipse Arrowhead framework supports QoS-aware failover and substitution mechanisms in practice (Section VI). In all, the results demonstrate that the proposed QoS-aware service model integrated with the layered DT-based composition architecture enable secure, interoperable, and composable abstractions, thereby decoupling cyber functionality from hardware constraints. These results are discussed in greater detail in Section VII.

II. Challenges and Problem Analysis

A. Research Challenges

Designing proxy-based DTs for autonomous industrial mobile machines requires addressing several intertwined challenges spanning interoperability, security, and composability. Here, we list such challenges identified in the literature:

1) Many industrial DT realizations remain hardware-centric and closely tied to proprietary vendor platforms, which inhibits interoperability, reuse, and evolution over long machine lifecycles in heterogeneous IIoT and OT environments [7], [16].
2) Industrial mobile machines operate in safety-critical settings where both sensing and actuation must be exposed digitally, making secure and dependable bidirectional cyber-physical interaction a fundamental requirement. In this context, DT-based security architectures highlight both the opportunities and the need for explicit security-by-design in industrial automation [15].
3) Fleet-level functionality (e.g., mission coordination, supervision, and safety monitoring) requires composing multiple machine-level twins, yet integration and composition challenges are well documented when engineering DT systems-of-systems [17].
4) Purpose-oriented DT compositions calls for a software-centric proxy abstraction that supports dynamic discovery, orchestration, and reconfiguration of twin services. Here, distributed iDTs as proxies provide an architectural basis for such composability and flexibility [5].
5) Existing approaches focus primarily on functional composition and availability in digital twin and CPS orchestration, and lack mechanisms to reason about quality differences among functionally equivalent services. While QoS-aware adaptation has been studied in service-based systems, it is not integrated into composable DT architectures, limiting the ability to support resilience through controlled degradation [5], [9], [17].

B. Fictitious Use Case

Consider an underground hard-rock mine where drill-and-blast operations are conducted as part of the production cycle. Following a blast event, the production area becomes temporarily inaccessible due to hazardous gases, unstable ground conditions, and potential rockfall hazards. Regulatory requirements mandate inspection and hazard verification before personnel re-entry, introducing delays that negatively impact production continuity and equipment utilisation. Such challenges are recognised in industrial cyber-physical production systems and digital twin-enabled environments [14], [20].

To address this scenario, a coordinated fleet of autonomous and tele-operable mobile robots is deployed to perform post-blast inspection and initial recovery operations. The fleet comprises heterogeneous platforms such as inspection robots, wheel-laoders, LHDs, and mobile sensing units, each equipped with multi-modal sensing capabilities (e.g., LiDAR, radar, gas sensors, and vision systems). The integration of such heterogeneous systems across operational technology (OT) and information technology (IT) domains introduces significant interoperability challenges [7], [16].

Following the concept of distributed Digital Twins (DTs) as proxies [5], each physical machine M_i is represented by an individual DT (iDT), denoted iDT_i, acting as a software proxy encapsulating the capabilities of the machine, e.g., sensing, autonomous task execution, and tele-remote control capabilities. Here, the concept of DTs as proxies, enables composability and flexibility. The iDT exposes its capabilities as standardized, discoverable services, abstracting vendor-specific implementations and enabling interoperable integration across heterogeneous platforms [16].

The inspection mission is realized through composite DTs (cDTs) orchestrating services from multiple iDTs. The cDTs are organised into two layer, where the lower-layer provides:

$\mathbf{cDT}_1$: Autonomous exploration and mapping
(using one or more autonomous mobile robots)
$\mathbf{cDT}_2$: Real-time gas concentration monitoring
(using one or more mobile sensing units)
$\mathbf{cDT}_3$: Detection and classification of hazards
(e.g., loose rock, misfires detected by mobile robots)
$\mathbf{cDT}_4$: Selective material handling and clearance
(using wheel-loaders and/or LHDs)
$\mathbf{cDT}_5$: Tele-remote intervention for manual operation
(i.e., controlling autonomous robots)

And the upper-layer provides:

cDT$_a$: Inspection and recovery
(based on cDT$_1$, cDT$_3$, cDT$_4$, cDT$_5$)
cDT$_b$: Hazard monitoring and access control
(based on cDT$_2$, cDT$_3$)

These upper-layer services reuse the same underlying capabilities, enabling independent and flexible composition. Runtime discovery and orchestration enable system-of-systems integration and dynamic binding of sensing, actuation, and control services across machines, supporting adaptation to environmental changes and partial failures [17].

Operations occur in connectivity-constrained environments, requiring edge-local execution of autonomy and safety functions, while synchronising selected data with supervisory systems when connectivity is available. iDTs ensure service availability despite temporary machine unavailability, aligning with distributed industrial architectures and secure DT-based control systems [11], [15].

This use case highlights key requirements for proxy-based DTs of industrial mobile machines:

- **Interoperability:** abstraction of heterogeneous machines into uniform, service-based iDTs,
- **Composability:** dynamic formation of mission-specific cDTs through orchestration,
- **Secure interaction:** controlled access to sensing and actuation services in safety-critical environments,
- **Resilience:** continued operation under intermittent connectivity through edge-local autonomy,
- **Scalability:** flexible inclusion or substitution of machines during mission execution.

Overall, the scenario demonstrates how proxy-based DTs enable coordinated, safe, and efficient operation of heterogeneous industrial robotic fleets [6].

C. Problem Analysis

The above use case highlights several architecture drivers that the conceptual architecture must address.

P1 – Heterogeneous Machine Integration. The fleet is composed of proprietary platforms with different sensing and control interfaces. A uniform cyber representation is required to abstract vendor-specific implementations into interoperable machine models.

P2 – Exposure of Cyber-Physical Capabilities. Each machine must expose its sensing outputs, autonomy task execution functions, and tele-remote operation capabilities through standardised, discoverable interfaces to enable composition and supervisory control.

P3 – Secure Bidirectional Interaction. Safety-critical environments require secure exchanges between supervisory layers and machines, enforcing authentication, authorisation, and policy controls for all sensed data and control commands.

P4 – Fleet-Level Coordination and Composition. Mission coordination, traffic management, and multi-machine supervision require runtime discovery and dynamic orchestration of distributed machine capabilities, rather than static point-to-point integrations.

P5 – Resilience Under Intermittent Connectivity. Connectivity constraints in underground operations necessitate edge-local autonomy and mechanisms for graceful degradation when central connectivity is disrupted.

P6 – Lifecycle Flexibility and Scalability. Architectures for mining automation must allow incremental evolution of assets over long lifecycles without disrupting existing supervisory or coordination functions.

III. Conceptual Architecture

This section presents the architecture for proxy-based Digital Twins (DTs) of autonomous industrial mobile machines. The novelty of the architecture lies in its specialization and operationalization of the proxy-based DT concept for industrial mobile machines, extended with a QoS-aware service model to support resilient operation under degraded conditions.

The architecture follows the concept of distributed DTs as proxies [5] and is grounded in the service-oriented and secure integration principles of Eclipse Arrowhead [11]. Its purpose is to provide a software-centric, composable, interoperable, and quality-aware abstraction of heterogeneous machines in safety-critical industrial environments.

A. Architectural Overview

The architecture separates the industrial environment into two interacting domains: the Physical Space (PS), where physical machines operate, and the Virtual Space (VS), where their corresponding digital representations execute as software systems. In the PS, mobile machines such as LHD vehicles, inspection robots, and sensing units perform sensing, actuation, and autonomous tasks. In the VS, their proxy-based digital counterparts expose these capabilities as discoverable and composable services. The interaction between PS and VS is enabled through service-oriented interfaces, allowing secure data exchange and control. This separation enables interoperability, flexibility, and scalable deployment across heterogeneous industrial assets.

B. Individual DTs as Proxies

Each physical machine M_i is represented by an Individual DT (iDT), denoted as iDT$_i$. The proxy relation is defined as:

$$\phi : M_i \rightarrow \mathrm{iDT}_i \tag{1}$$

The iDT encapsulates externally relevant capabilities of the machine, including sensing, autonomous task execution, tele-remote control, and lifecycle management functions.

This proxy abstraction isolates service consumers from hardware-specific implementations, thereby supporting interoperability, modularity, and lifecycle flexibility in heterogeneous industrial environments.

C. Service-Oriented Capability Representation

The capabilities of each iDT are exposed as services in accordance with Service-Oriented Architecture (SOA) principles [18]. Let $S_i = \{s_{i1}, s_{i2}, \ldots, s_{in}\}$ denote the service set

of iDT_i. To support resilience and informed service selection, each service is extended with Quality-of-Service (QoS) attributes. A service is therefore defined as:

$$s_{ij} = (f_{ij}, Q_{ij}) \tag{2}$$

where f_{ij} denotes the functional capability, and Q_{ij} represents a vector of QoS attributes:

$$Q_{ij} = \{q_1, q_2, \ldots, q_n\} \tag{3}$$

Typical QoS attributes include sensing accuracy, latency, reliability, and data freshness. This representation enables distinguishing between functionally equivalent services that differ in quality. This QoS-extended service model forms the basis for QoS-aware substitution and orchestration, defined in the following subsection.

D. QoS-Aware Service Model and Substitution

In heterogeneous CPS environments, redundant services are often only partially equivalent. To formalize substitution, we define a partial order over services:

$$s_a \preceq s_b \iff f_a = f_b \wedge Q_a \leq Q_b \tag{4}$$

This relation captures that service s_b dominates s_a if it provides the same functionality with equal or better QoS. To support runtime decision-making, a utility function is defined:

$$U(s) = \sum_{k=1}^{n} w_k \cdot q_k \tag{5}$$

where w_k are context-dependent weights reflecting mission priorities (e.g., prioritizing accuracy over latency in safety-critical sensing).

The weighted-sum formulation is chosen for its computational simplicity and suitability for real-time decision-making in resource-constrained edge environments. While more advanced multi-objective optimization methods exist, this formulation provides a transparent and tunable mechanism aligned with mission-level priorities.

Given a set of available services S_{avail}, where S_{avail} denotes the set of currently available candidate services, the orchestrator selects:

$$s^* = \arg\max_{s \in S_{avail}} U(s) \tag{6}$$

This enables selection among partially equivalent services and supports *QoS-aware service selection*, where lower-quality services are used when higher-quality ones are unavailable. The approach is consistent with QoS-aware service selection and adaptive systems [9], [10], [13]. The computational complexity of service selection is linear in the number of candidate services, making the approach suitable for real-time orchestration in dynamic industrial environments.

E. Composite DTs for Purpose-Oriented Composition

Composite Digital Twins (cDTs) are formed by orchestrating services from multiple iDTs to realize mission-specific functionality. Given a purpose P_k, a cDT is defined as:

$$\mathrm{cDT}_{P_k} = \bigcup_{j=1}^{m} S_j^{P_k} \tag{7}$$

where $S_j^{P_k} \subseteq S_j$ denotes the subset of services relevant to the purpose.

Unlike monolithic twins, cDTs are dynamically composed and are not tied to a single physical asset. This enables flexible system-of-systems integration and reuse of capabilities across missions.

F. Runtime Composition and Orchestration

Fleet-level and mission-level behaviour is achieved through runtime orchestration of distributed services into cDTs. The orchestration layer selects services based on mission purpose, availability, context, and QoS characteristics. This layer provides an orchestration mechanism through core components supporting discovery, orchestration, authorization, and lifecycle management. The Service Registry enables discovery of iDT and cDT services, the Orchestration System performs runtime binding and QoS-aware service selection, the Authorization System enforces secure service access, and the DT Deployment Support manages instantiation and lifecycle of DT services.

The orchestration mechanism builds on the QoS-aware service model defined in Section III-D, enabling runtime selection among partially equivalent services. This allows resilience through controlled degradation, i.e., the ability to maintain functionality by selecting lower-quality but functionally equivalent services when necessary.

The orchestration layer supports deployment across edge-local and cloud-connected environments. iDTs are typically deployed near physical machines to ensure low latency and local autonomy, while higher-level cDTs coordinate across machines and sites. This supports low-latency control, continued operation under intermittent connectivity, scalable coordination across fleets, and selective synchronization with supervisory systems.

IV. Proof-of-Concept

A. PoC Scenario

This scenario operationalises the fictitious use case introduced in Section II-B, instantiating the abstract fleet and service structure as concrete iDT instances within the PoC environment. To reflect realistic fleet operation, multiple instances of selected iDT types are instantiated. Specifically, two instances are used for inspection robots (iDT_{1a}, iDT_{1b}), gas sensing units (iDT_{2a}, iDT_{2b}), and LHD vehicles (iDT_{3a}, iDT_{3b}), while a single tele-remote station (iDT_4) provides centralized supervision.

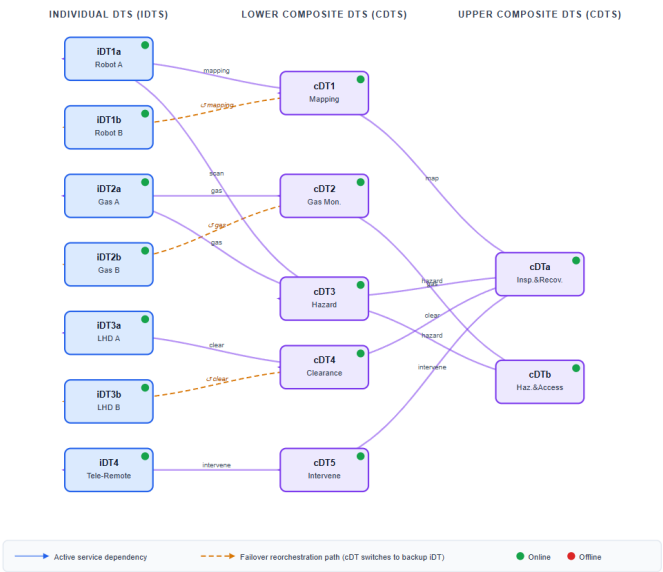

Fig. 1. Layered composition of individual DTs (iDTs) and composite DTs (cDTs) in the PoC scenario. The figure shows the iDT and cDT instances, along with their service interactions.

The PoC applies the QoS-aware service selection model defined in Section III-D. The PoC includes controlled degradation and failure scenarios, where alternative iDT providers with different QoS characteristics are dynamically selected during operation. In the PoC, QoS attributes are instantiated using measurable metrics such as sensing variance (accuracy), service response latency, and availability, which are mapped to the utility function defined in Section III-D.

To support composability and avoid duplication of functionality, the PoC adopts a two-layer Composite Digital Twin (cDT) architecture. Lower-layer cDTs provide reusable, shared capabilities, while higher-layer cDTs implement mission-specific logic. The roles and interactions of iDT instances and cDTs are summarised in Table I and illustrated in Figure 1.

At the lower layer, cDT1 performs autonomous exploration and SLAM-based mapping by aggregating data from the inspection robot iDT instances (iDT1a, b), while cDT2 provides real-time gas concentration monitoring from the gas sensing iDT instances (iDT2a, b). cDT$_3$ fuses robot scan and gas data to detect and classify hazards, maintaining a quality-aware environment model based on QoS characteristics of the incoming data streams. cDT4 orchestrates debris clearance using the LHD iDT instances (iDT3a, b), and cDT$_5$ provides tele-remote intervention and manual override via iDT$_4$.

After a blast event, all sensing iDT instances provide data consumed by the lower-layer cDTs. When a sensing source becomes unavailable or degraded, alternative iDT services are selected according to the QoS-aware utility function defined in Section III-D. At the upper layer, cDT$_a$ coordinates the post-blast inspection and recovery mission by orchestrating task allocation across inspection, clearance, and intervention capabilities (uses cDT$_1$, cDT$_3$, cDT$_4$, cDT$_5$). In parallel, cDT$_b$ performs hazard assessment and access control, integrating gas monitoring and hazard detection to determine safe re-entry conditions and gate access (uses cDT$_2$, cDT$_3$). The tele-remote iDT enables human supervision when required.

Here, iDTs do not interact directly. Instead, all coordination is performed through the cDTs, which dynamically compose and orchestrate services across multiple iDT instances. This orchestration extends from availability-based selection to QoS-aware service selection, enabling resilience through controlled degradation rather than binary failover. This demonstrates the scalability and composability of proxy-based DTs [5], [17].

The layered structure of the PoC, including QoS-aware service composition and substitution across iDT and cDT layers, is summarised in Table I.

B. Implementation

The iDTs are implemented as proxy services in the Virtual Space. The Eclipse Arrowhead core components support the PoC through service registration, secure access control, and runtime orchestration. Each iDT is created to capture sensing properties, control commands, and connectivity information. The resulting proxy is then registered and exposed as a service endpoint, making it available for composition into cDTs. This is consistent with the software-centric and modular approach emphasized in the earlier sections. The Physical Space is emulated to enable controlled and repeatable experimentation. However, the system architecture, communication patterns, and service interactions reflect realistic industrial deployments based on mining automation scenarios.

The implementation is available on GitHub under the MIT license [8]. The PoC setup includes seven iDT instances and seven composed services provided by CDTs, reflecting a small-scale industrial fleet. This setup allows evaluation of orchestration behaviour under realistic multi-entity conditions.

V. Evaluation

To analyse the conceptual properties of the QoS-aware service model, we conduct controlled experiments using generalized sensing and actuation capabilities derived from the PoC scenario, enabling exploration of QoS trade-offs and degradation behaviour. These capabilities are abstracted into representative iDT services with varying QoS characteristics. The evaluation focuses on (i) trade-off behaviour under varying mission priorities and (ii) resilience under controlled degradation. The evaluation addresses two research questions: (RQ1) how QoS-aware selection influences service choice under varying mission priorities, and (RQ2) to what extent QoS-awareness can improve system resilience under degradation.

All results are obtained from repeated randomized experiments with a configurable number of runs and a seed to ensure reproducibility.

A. Experimental Setup

A discriminative scenario with clear QoS diversity and strong trade-offs between accuracy, latency, and reliability is used for the evaluation. Each service is characterised by a QoS vector including accuracy, latency, and reliability.

TABLE I
LAYERED ARCHITECTURE WITH MULTIPLE iDT AND cDT INSTANCES IN THE PoC SCENARIO

Layer	Component	Functions and Usage
Physical / iDT	$\text{iDT}_{1a,b}$ (Inspection Robots)	SLAM, hazard detection, localisation, navigation
	$\text{iDT}_{2a,b}$ (Gas Sensors)	Gas monitoring, alerts (QoS: accuracy, latency)
	$\text{iDT}_{3a,b}$ (LHD Vehicles)	Material handling, clearance, autonomous tramming
	iDT_4 (Tele-remote)	Tele-operation, override, supervision
Lower cDT	cDT_1 (Mapping)	SLAM-based mapping, QoS-aware robot selection (uses $\text{iDT}_{1a,b}$)
	cDT_2 (Gas Monitoring)	Gas monitoring, threshold alerting, QoS-aware sensor selection (uses $\text{iDT}_{2a,b}$)
	cDT_3 (Hazard Detection)	Hazard detection/classification, fuses robot and gas data (uses iDT_{1a}, iDT_{2a})
	cDT_4 (Clearance)	Debris clearance, QoS-aware vehicle selection (uses $\text{iDT}_{3a,b}$)
	cDT_5 (Intervention)	Tele-remote intervention and manual override (uses iDT_4)
Upper cDT	cDT_a (Inspection & Recovery)	Inspection, task allocation, QoS-aware coordination (uses cDT_1, cDT_3, cDT_4, cDT_5)
	cDT_b (Hazard Monitoring & Access)	Hazard assessment, safe-access decisions, gate control (uses cDT_2, cDT_3)

Utility is computed using the weighted sum defined in Section III-D, with weights varied systematically to explore trade-offs. For example, high-accuracy providers correspond to advanced sensing units (e.g., LiDAR-equipped inspection robots), while low-latency providers represent degraded sensing alternatives.

Two selection strategies are compared: ***(A) QoS-aware (proposed):*** selects services based on utility maximisation, and ***(B) Availability-based (baseline)*:** a commonly used availability-based selection strategy, which selects the first available service without considering QoS differences.

Table II lists the parameters used in the experiments. Two service providers are modelled with deliberately non-overlapping quality profiles: Provider A offers high accuracy (90–99%), high latency (55–95,ms), and high reliability (92–99%), whereas Provider B offers low accuracy (38–62%), low latency (2–9,ms), and moderate reliability (52–76%). Each run samples provider profiles uniformly within these ranges. here, the QoS-aware selection algorithm weights accuracy most heavily ($w_{\text{acc}} = 0.50$), with equal shares for latency and reliability ($w_{\text{lat}} = w_{\text{rel}} = 0.25$), and applies a utility failover threshold of 0.50, a switch-back hysteresis of 0.05, and an availability failover threshold of 0.12.

TABLE II
PARAMETERS FOR THE EXPERIMENTS USING THE PoC SCENARIO

Parameter	Value
Provider profiles (sampled per run)	
A: accuracy / latency (ms) / reliability	$[0.90, 0.99]$ / $[55, 95]$ / $[0.92, 0.99]$
B: accuracy / latency (ms) / reliability	$[0.38, 0.62]$ / $[2, 9]$ / $[0.52, 0.76]$
Nominal QoS	
A: accuracy / latency (ms) / reliability	0.97 / 18 / 0.995
B: accuracy / latency (ms) / reliability	0.68 / 5 / 0.82
Timing & degradation	
Duration / episodes / recovery (s)	200 / 3 / 10
Degrade rate (acc s^{-1}) / onset (s)	$[0.05, 0.18]$ / $[12, 20]$
Degrade window / fail window / gap (s)	$[8, 16]$ / $[8, 14]$ / $[5, 12]$
Selection algorithm	
w_{acc} / w_{lat} / w_{rel}	0.50 / 0.25 / 0.25
Util. threshold / hysteresis	0.50 / 0.05
Availability threshold	0.12

B. QoS Trade-off Analysis

We first analyse how the model behaves under varying mission priorities by sweeping the accuracy weight $w_{\text{acc}} \in [0, 1]$ while distributing the remaining weight equally between latency and reliability.

Figure 2 shows the utility of two representative providers as a function of w_{acc}. Provider A offers higher accuracy and reliability at the cost of higher latency, while Provider B offers lower latency but reduced accuracy and reliability. A clear crossover point is observed around $w_{\text{acc}} \approx 0.30$, where the preferred provider changes.

These results demonstrate that the QoS-aware model produces *interpretable and predictable decision policies*, where changes in mission priorities lead to well-defined shifts in selected services. Note that the transition is not gradual but reflects discrete provider selection.

C. Controlled Degradation and Resilience

We next evaluate system behaviour under controlled degradation. In this scenario, multiple degradation windows are introduced, during which the quality of the primary service deteriorates. The system must then select alternative services to maintain functionality.

Figure 3 (top) shows the utility over time for both the QoS-aware and availability-based approaches. Outside degradation periods, both approaches achieve similar performance. However, during degradation windows, the baseline exhibits sharp drops in utility, in some cases falling below 0.3, while the QoS-aware approach maintains significantly higher utility, typically above 0.75.

The bottom part of Figure 3 shows the utility advantage of the QoS-aware approach relative to the baseline. The advantage is negligible outside degradation periods, but becomes substantial during degradation, with median gains of 0.15–

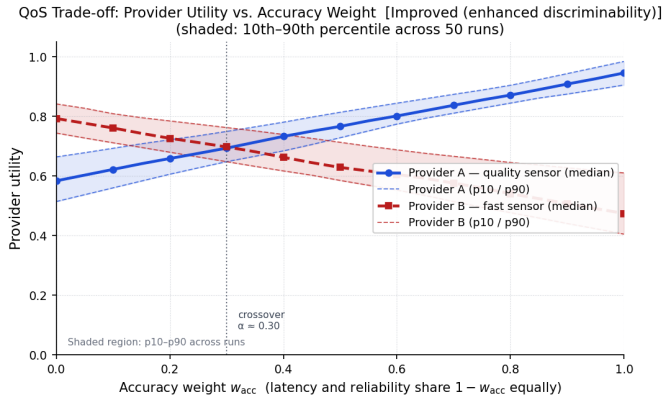

Fig. 2. QoS trade-off: provider utility vs. accuracy weight. The crossover point indicates a shift in the preferred provider. Shaded regions represent the 10th–90th percentile across runs.

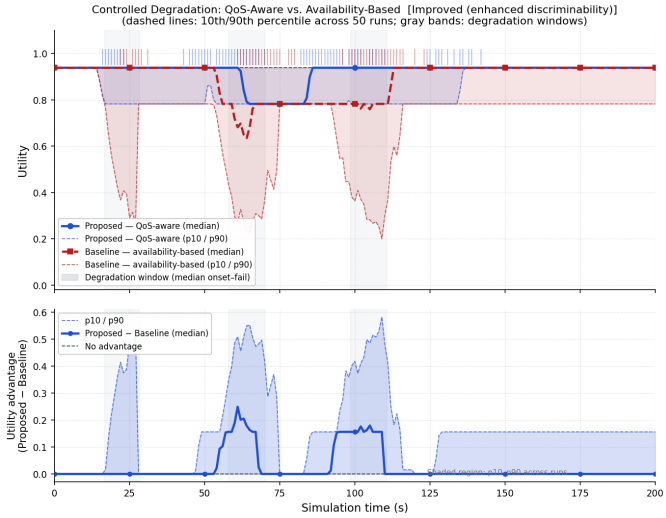

Fig. 3. Controlled degradation scenario. **Top:** Utility over time for QoS-aware and availability-based selection. **Bottom:** Utility advantage (QoS-aware minus baseline). Shaded regions denote degradation windows; dashed lines indicate 10th–90th percentiles.

0.25 and peaks exceeding 0.5. During degradation periods, QoS-aware selection maintains utility above 0.75, while the baseline drops below 0.3, corresponding to a relative improvement of more than 100% in worst-case scenarios.

These results confirm that the proposed model enables *controlled degradation*. Rather than experiencing abrupt performance collapse, the system adapts by selecting lower-quality but functionally equivalent services.

D. Discussion

The evaluation highlights three key properties of the proposed approach:

- **Interpretable trade-offs:** The QoS-aware model yields clear and predictable decision boundaries, allowing system behaviour to be tuned via mission-level priorities.
- **Graceful degradation:** The system maintains higher utility under adverse conditions, avoiding the brittle behaviour of availability-based selection.
- **Context-dependent benefits:** QoS-awareness provides significant advantages during degradation while introducing no penalty under nominal conditions.

The evaluation is based on abstracted service providers and an emulated physical environment, enabling controlled and repeatable experimentation of QoS trade-offs and degradation behaviour. While this approach allows systematic analysis of the proposed model, it does not capture all aspects of real industrial deployments, such as hardware-induced latency, network variability, and system integration constraints.

Furthermore, the baseline comparison is limited to availability-based selection, and additional comparisons with more advanced orchestration strategies could provide further insights. Practical deployment also requires that such mechanisms introduce minimal runtime overhead. This aspect is analysed in the following subsection.

VI. Arrowhead Support for QoS-Aware Failover

This section evaluates how the Eclipse Arrowhead framework supports the implementation of the proposed QoS-aware service model in practical deployments.

A. Service Discovery and Multi-Endpoint Availability

A fundamental requirement for QoS-aware failover is the ability to discover and access multiple functionally equivalent service providers at runtime. The Arrowhead Service Registry explicitly supports this capability by maintaining a catalogue of available service instances and enabling filtering based on service definitions and metadata. The *serviceDiscovery* functionality allows consumers to retrieve multiple candidate providers for a given service, rather than a single endpoint [3]. This aligns with the requirement in Section III-D that the orchestrator operates over a set of available services S_{avail}, enabling selection among alternative providers.

B. Dynamic Orchestration and Runtime Binding

The Arrowhead Orchestration System provides mechanisms for dynamic service binding at runtime. In particular, the *Dynamic Service Orchestration* functionality supports matching between service consumers and multiple candidate providers, enabling flexible runtime configuration of service interactions [2]. This capability is consistent with the compositional model presented in this paper, where Composite Digital Twins (cDTs) dynamically bind services from multiple Individual Digital Twins (iDTs) based on mission context.

C. QoS Support and Evaluation Mechanisms

Arrowhead includes explicit support for QoS-related evaluation through the *Device QoS Evaluator* support system. This component enables filtering and ranking of service providers based on KPIs such as latency, reliability, and other measurable attributes [1]. Furthermore, the orchestration configuration supports QoS-related considerations (e.g., QoS-enabled orchestration flags), allowing QoS metrics to influence service

selection. This aligns with the QoS vector Q_{ij} and utility-based selection model defined in Section III-D.

However, the Arrowhead framework does not prescribe a specific utility function or decision model. Instead, it provides the mechanisms for retrieving QoS data and candidate services, leaving the final selection logic to the application or orchestration layer.

D. Conclusion of Assessment

From the analysis above, Arrowhead satisfies the key structural requirements for QoS-aware failover, i.e., support of (a) multiple service endpoints, (b) runtime binding, and (c) QoS data availability. Together, these capabilities enable an application-level orchestrator to implement the selection function defined in Equation 6.

Thus, the Arrowhead framework supports *multi-provider service exposure*, meaning that an application system can obtain a set of candidate endpoints and switch between them at runtime. This makes it possible to implement immediate failover by selecting an alternative provider when the currently used service degrades or becomes unavailable.

Despite providing the necessary infrastructure, Arrowhead does not implement QoS-aware failover as a built-in, automatic mechanism. Specifically, the framework does not enforce continuous monitoring and automatic switching based on QoS thresholds. Nor is the selection policy (e.g., utility function, weighting of QoS attributes) defined within the core systems. This means that failover behaviour must be implemented explicitly in the orchestration logic or in higher-level application components (e.g., cDTs).

VII. Conclusions and Future Work

This paper presented a service-oriented architecture for proxy-based Digital Twins (DTs) of autonomous industrial mobile machines, grounded in distributed DT concepts and Eclipse Arrowhead. By abstracting machines into individual DTs (iDTs) exposing standardized services, the approach enables dynamic composition into purpose-oriented composite DTs (cDTs) for fleet- and mission-level functionality.

The results show that proxy-based iDTs support decoupling from hardware-specific implementations while enabling secure, discoverable, and composable services, thus validating the feasibility of flexible and scalable system-of-systems integration in safety-critical environments. The evaluation further demonstrated that QoS-aware orchestration maintains significantly higher system utility under degradation scenarios compared to availability-based selection.

The analysis of the Eclipse Arrowhead framework indicates that it provides the necessary architectural mechanisms to support the proposed QoS-aware service model. In particular, the framework enables discovery of multiple service providers, dynamic orchestration and runtime binding, and access to QoS-related metrics through supporting systems. However, Arrowhead does not implement QoS-aware failover as a built-in mechanism. Instead, the selection logic and failover behaviour must be realized within the application or orchestration layer.

Future work includes validation in real industrial settings to assess performance and robustness, as well as the development of more advanced security mechanisms. Further work also includes verifying and validating Eclipse Arrowhead for QoS-aware failover beyond the Proof-of-Concept described in this paper. Further research is also needed on adaptive orchestration strategies and formal methods for safe composition.

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